Array Practice Questions (C++)

Question 1: Adjacent Difference Count

You are given an array of N integers and an integer K. Count the number of adjacent pairs (A[i], A[i+1]) such that the absolute difference between them is exactly K.

In other words, count all indices i ($0 \le i < N-1$) for which:
$|A[i] - A[i+1]| = K$

Print only the total count.

Example

Input:
N = 7
K = 3
Array = {5, 2, 6, 9, 12, 10, 7}

Output:
4

Explanation:
|5-2|=3 ✓
|2-6|=4 ✗
|6-9|=3 ✓
|9-12|=3 ✓
|12-10|=2 ✗
|10-7|=3 ✓
Total = 4

Question 2: Equal Adjacent Difference

You are given an array of N integers. For every adjacent pair, compute the difference:
$D[i] = A[i+1] - A[i]$

Determine which difference value occurs the maximum number of times among all adjacent pairs, and print its frequency.

Example

Input:
N = 8
Array = {3, 5, 7, 10, 13, 16, 18, 20}

Output:
4

Explanation:
Differences: 2, 2, 3, 3, 3, 2, 2
Difference 2 appears 4 times.
Difference 3 appears 3 times.
Output = 4